Specification

ZR/YC-0545 S3

97 Jinye 2nd Road,Xi'an Shaanxi 710119 China
Tel: +86 029—68590655 68590656 68590678(sales)
Fax: +86 029—68590676
info@sinofuse.com
http://www.sinofuse.com

DC500V 160-630A

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DC500V 160-630A

Basics

- Conform to: IEC 60269，UL248
- Rated Voltage: DC500V
- Rated Current: 160A～630A
- Utilization Category: aBat
- Breaking Capacity: DC 50kA
- RoHS compliant,UL,TUV,CE certified

This product is a fuse designed for partial range protection, specifically for battery pack protection in energy storage systems. It features high I²t characteristics and is suitable for short-circuit protection in energy storage battery systems, as well as other suitable circuit systems.

Note: All test data in this specification sheet was obtained under standard test conditions, with test parameters and wiring in accordance with IEC 60269.

No.	Part Number	Size	Rated Current A	I²t（A²s）b		I²t（A²s）c		Loss		Weight（g）	Min Package	Max Package
				Prearc	Clearing	Prearc	Clearing	0.5In	In			
1	RS08R-LGH 160A	00	160	52400	71400	34900	72000	5.07	27.9	200±15	15	60
2	RS08R-LGH 200A		200	103000	140000	68300	141000	5.7	31.5			
3	RS08R-LGH 250A		250	170000	232000	113000	233000	7.01	38.9			
4	RS08R-LGH 315A		315	302000	412000	201000	415000	8.45	47.2			
5	RS08R-LGH 350A		350	422000	575000	281000	580000	8.91	49.9			
6	RS08R-LGH 400A		400	562000	766000	373000	770000	10.2	57.2			
7	RS08R-LGH 450A		450	809000	1100000	538000	1110000	10.9	61.5			
8	RS08R-LGH 500A		500	1100000	1500000	732000	1510000	11.6	65.9			
9	RS08R-LGH 550A		550	1440000	1960000	956000	1970000	12.4	70.5			
10	RS08R-LGH 630A		630	1820000	2480000	1210000	2500000	14.7	84.6			

Note: a. Loss value measured based on RS08R-LGQ test data

b. I²t data based on 20.8kA at DC508V, time constant 13.9ms

c. I²t values based on I1 and I2 test data at DC508V, time constant 10-15ms.

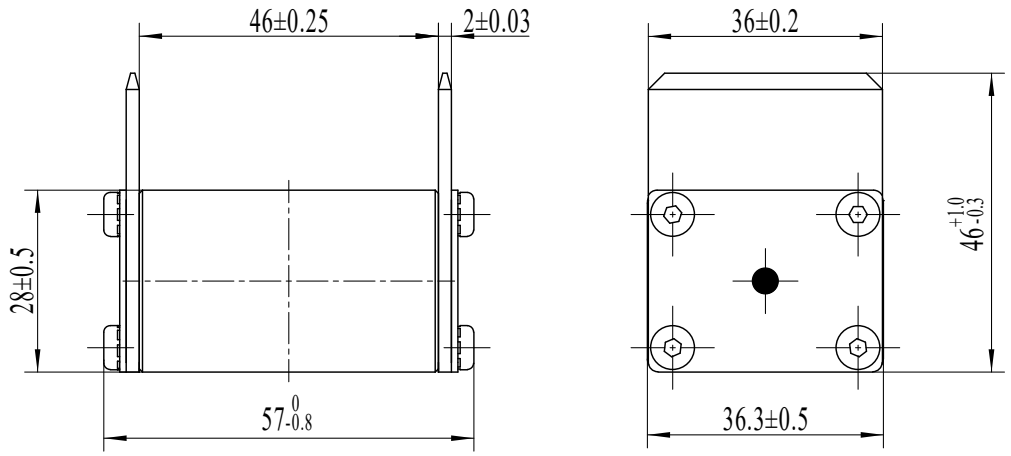
RS08R-LGH Mounting

Mounting

Mounted with MSD

Outline and Dimensions (mm)

RS08R-LGH Mounting Dimension



DC500V 160-630A

No.	Part Number	Size	Rated Current A	I²t（A²s）b		I²t（A²s）c		Loss		Weight（g）	Min Package	Max Package
				Prearc	Clearing	Prearc	Clearing	0.5In	In			
11	RS08R-LGHB 160A	00	160	52400	71400	34900	72000	5.07	27.9	210±15	15	60
12	RS08R-LGHB 200A		200	103000	140000	68300	141000	5.7	31.5			
13	RS08R-LGHB 250A		250	170000	232000	113000	233000	7.01	38.9			
14	RS08R-LGHB 315A		315	302000	412000	201000	415000	8.45	47.2			
15	RS08R-LGHB 350A		350	422000	575000	281000	580000	8.91	49.9			
16	RS08R-LGHB 400A		400	562000	766000	373000	770000	10.2	57.2			
17	RS08R-LGHB 450A		450	809000	1100000	538000	1110000	10.9	61.5			
18	RS08R-LGHB 500A		500	1100000	1500000	732000	1510000	11.6	65.9			
19	RS08R-LGHB 550A		550	1440000	1960000	956000	1970000	12.4	70.5			
20	RS08R-LGHB 630A		630	1820000	2480000	1210000	2500000	14.7	84.6			

Note: a. Loss value measured based on RS08R-LGQ test data

b. I²t data based on 20.8kA at DC508V, time constant 13.9ms

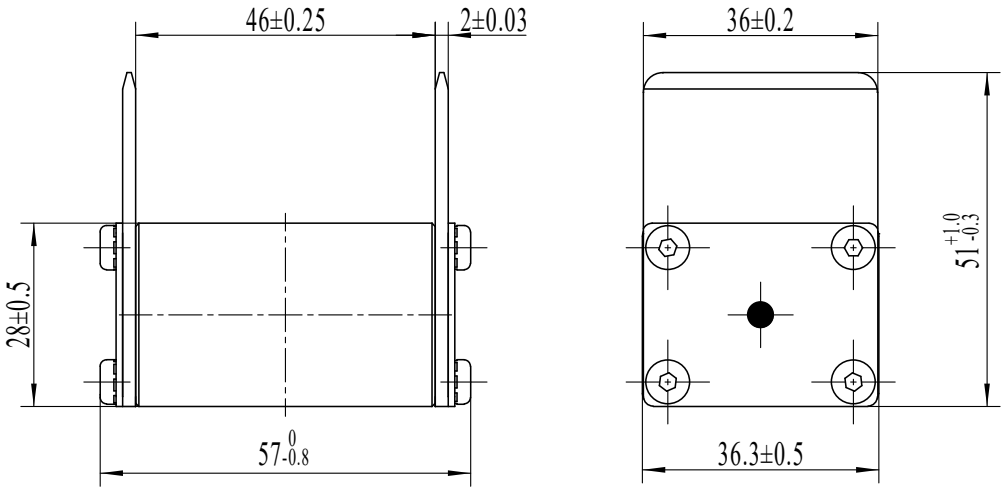
c. I²t values based on I1 and I2 test data at DC508V, time constant 10-15ms.

RS08R-LGHB Mounting

Mounting
Mounted with MSD

Outline and Dimensions (mm)

RS08R-LGHB Mounting Dimension



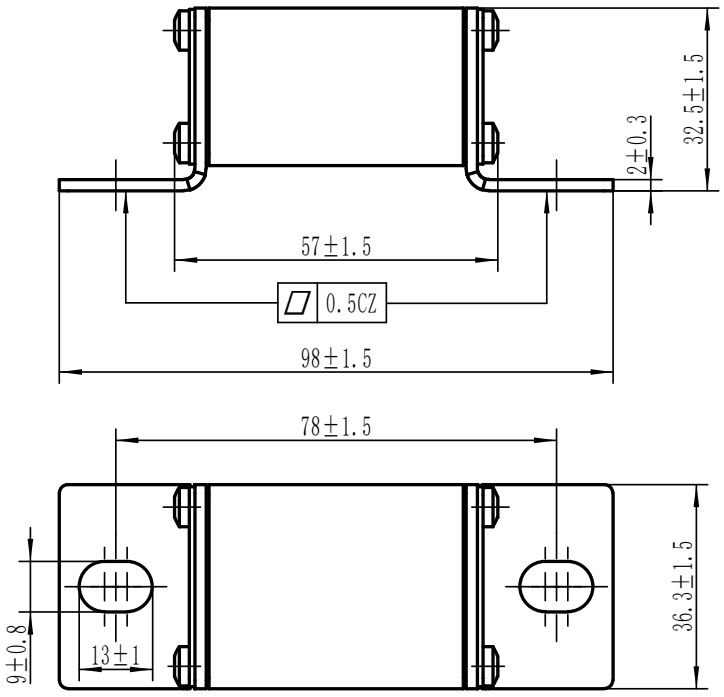
DC500V 160-630A

No.	Part Number	Size	Rated Current A	I²t （A²s）		Loss		Weight g	Min Packag	Max Packag	Note
				Prearc	Clearing	0.5In	In				
1	RS08R-LGQ 160A	00	160	34900	72000	5.07	27.9	210±15	12	48	Bolt M8 Torque 12±1N.m
2	RS08R-LGQ 200A		200	68300	141000	5.7	31.5				
3	RS08R-LGQ 250A		250	113000	233000	7.01	38.9				
4	RS08R-LGQ 315A		315	201000	415000	8.45	47.2				
5	RS08R-LGQ 350A		350	281000	580000	8.91	49.9				
6	RS08R-LGQ 400A		400	373000	770000	10.2	57.2				
7	RS08R-LGQ 450A		450	538000	1110000	10.9	61.5				
8	RS08R-LGQ 500A		500	732000	1510000	11.6	65.9				
9	RS08R-LGQ 550A		550	956000	1970000	12.4	70.5				
10	RS08R-LGQ 630A		630	1210000	2500000	14.7	84.6				

Note: The above I²t values based on I1, I2 test data at DC508V, time constant 10～15ms.

Outline and Dimensions (mm)

RS08R-LGQ Mounting Dimension



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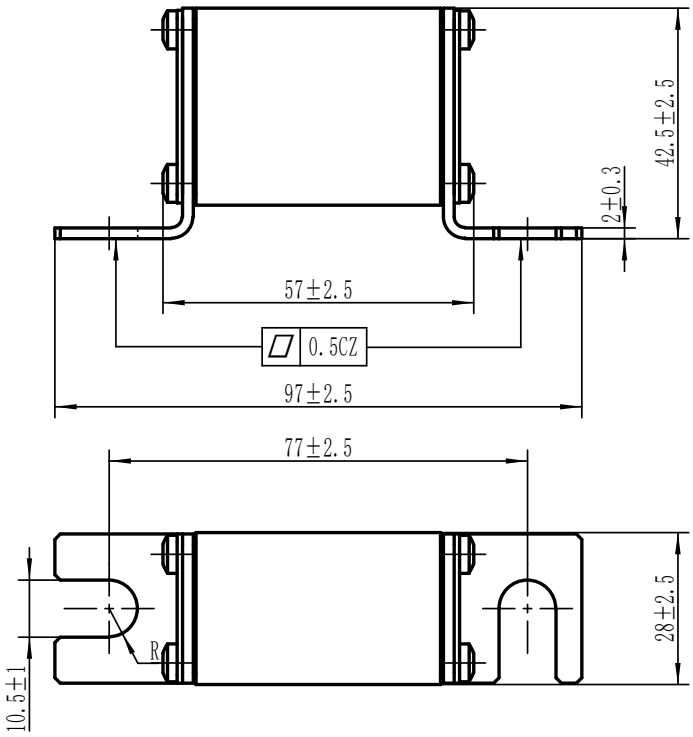
No.	Part Number	Size	Rated Current A	I²t （A²s）		Loss		Weight g	Note
				Prearc	Clearing	0.5In	In		
1	RS08R-LGN 160A	00	160	34900	72000	5.07	27.9	200±15	Bolt M10 Torque 21±2N·m
2	RS08R-LGN 200A		200	68300	141000	5.7	31.5		
3	RS08R-LGN 250A		250	113000	233000	7.01	38.9		
4	RS08R-LGN 315A		315	201000	415000	8.45	47.2		
5	RS08R-LGN 350A		350	281000	580000	8.91	49.9		
6	RS08R-LGN 400A		400	373000	770000	10.2	57.2		
7	RS08R-LGN 450A		450	538000	1110000	10.9	61.5		
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9	RS08R-LGN 550A		550	956000	1970000	12.4	70.5		
10	RS08R-LGN 630A		630	1210000	2500000	14.7	84.6		

Note: a. Loss value measured based on RS08R-LGQ test data

b. I²t values based on I1 and I2 test data at DC508V, time constant 10-15ms.

Outline and Dimensions (mm)

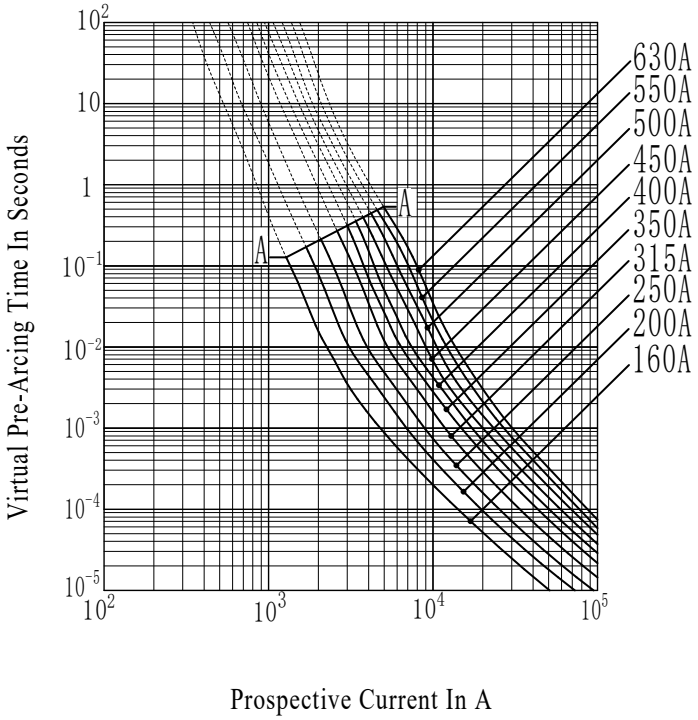
RS08R-LGN Mounting Dimension



DC500V 160-630A

Characteristic Curves

Time-current characteristic curve



Note: 1. 100ms or less in the curve is the equivalent pre-arcing time;
2.Time current curve tolerance ±15%
3.Min breaking capacity ⩾8In

Transport and Storage

Transport

Avoid rain/snow or mechanical damage during transportation

Storage

Storage temp: -40°C～120°C, Maximum 70% RH at 40°C;
Maximum 80% RH at 30°C; Maximum 90% RH at 20°C;
Package and Storage temp: -40°C～70°C, Maximum 90% RH ; no dewing.

Usage Condition

Normal Conditions

Correction is not required under normal conditions.
For other conditions, if they are within tolerable range, certain correction measures may be required. If conditions are beyond tolerable range, please consult our team for evaluation and testing
Long-term carrying current should not exceed 0.8 In

Ambient Temperature

Normal Range
-5℃～40℃

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Tolerable Range

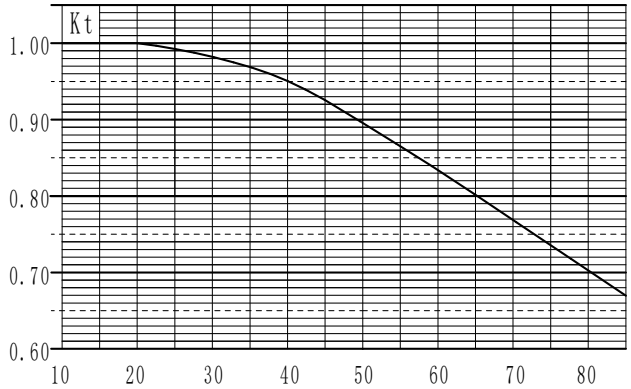
-40℃～85℃

Ambient temperature correction: operating below -5°C, resulting in longer pre-arc time under small overcurrent and slightly increased rated current.

If above 40°C, rated current is corrected as per factor -Kt.

Note 1: Kt value has considered safety margin during normal operation

Note 2: ambient temperature should last 1-2 hrs before it has significant impact on fuse



Altitude

Normal Condition

Below 2000m

Tolerable Condition

Below 4500m

Correction: higher altitude would affect insulation and dissipation, also changes air pressure.

a) For every 100m higher, fuse temperature rise increases by 0.1-0.5k

b) For every 100m higher, ambient temperature drops by 0.5k approximately

c) Normally for fuses in open environment, altitude condition is negligible

d) For closed environment, if ambient temperature inside remains almost stable under different altitude. If above 40°C, fuse should be derated. For every 1000m, rated current should be derated by 2%-5%

Note: for any series, larger rated fuse should use higher degrade %, and lower degrade % for smaller one.

Air Insulation Strength (Breakdown)

a) Air insulation reduces with higher altitude. For 2000-4500m, decreases by 12-15% for every 1000m, as per GB/T16935.1. Thus adjust clearing space.

b) Space between fuse terminals is often much larger than specified value in standard (GB/T16935.1).

c) User should consider altitude impact on spacing between fuse and other component, earthing etc.

Atmosphere

Normal Conditions

Clean atmosphere, maximum 50% RH at 40°C.

Higher RH is allowed when temperature is low, e.g. maximum 90% at 20 °C.

Moderate dewing may occur under temperature changes.

Tolerable Conditions

If dewing is minor, RH could be up to 95%.

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Vibration

This series of fuse has great withstand of vibration and shock and satisfies ISO16750-3 vehicle usage condition

For severe vibration application, please consult our team for evaluation and testing.

Pollution class

Grade 3 pollution withstand

Usage Condition

Normal Conditions

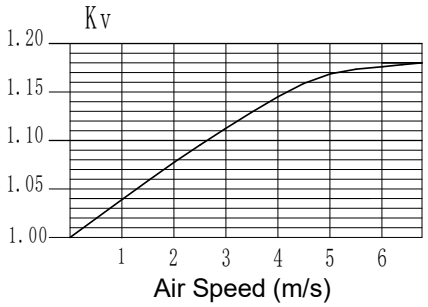
a) Fuse and fuse-holder should be mounted properly with specified torque and stress.

b) Contact of fuses must be securely connected. Contact resistance should not affect operation.

c) Fuse can be mounted in any orientation. If spring compression is adopted, make sure it is properly mounted to avoid harmful effect due to gravity or vibration.

Forced Air Cooling

Forced air cooling can be implemented to improve current carrying capacity of fuse.



Safety and Maintenance

a) Sufficient space must be ensured between installed fuses. Install insulation if necessary. This is to avoid possible inter-phase short circuit while replacing fuse.

b) Periodic maintenance of fuses includes removal of oxidation layer and dusts.

c) It is compulsory to replace all mechanically damaged fuses.

d) Unless permissive (e.g.fused load-switch), do not replace fuses while energized.

e) While servicing, fuse will not generate gas, dust, noise or others that may harm the environment

f) Metallic part of fuse can be recycled. Non-metal part can be crushed and treated as normal industry waste. It will not cause further pollution to the environment.